Shiping Yang

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EDUCATION

Simon Fraser University — **Vancouver, Canada**
Ph.D. student in Computing Science, Supervisor: Prof. Angel Xuan Chang — *09/2025 - 06/2029 (Expected)*

Beijing University of Posts and Telecommunications — **Beijing, China**
Master of Engineering in Computer technology — *09/2022 - 06/2025*

Beijing Language and Culture University — **Beijing, China**
Bachelor of Engineering in Computer Science — *09/2018 - 06/2022*

PUBLICATIONS

Distilling the Shape of Uncertainty: What On-Policy Distillation Learns and How to Predict It
Shiping Yang, *Chuanqi Tang, Hossein Hajimirsadeghi, Layla El Asri, Mohammad Amin Shabani, Angel X Chang* — ***Under Review***

Hallucination Self-Play: Bootstrapping Reinforced Detector via Evolved Generator
Shiping Yang, *Shining Liang, Weihao Liu, Wenbiao Ding, Linjun Shou, Lu Cheng, Angel X Chang* — ***COLM2026***

Quantifying and Improving the Robustness of Retrieval-Augmented Language Models Against Spurious Features in Grounding Data
Shiping Yang, *Jie Wu, Wenbiao Ding, ... , Hengyuan Zhang, Angel X Chang, Dongmei Zhang* — ***ACL2026 Main***

A New Dataset and Empirical Study for Sentence Simplification in Chinese
Shiping Yang, *Renliang Sun, Xiaojun Wan* — ***ACL2023 Main***

A New Benchmark and Reverse Validation Method for Passage-level Hallucination Detection
Shiping Yang, *Renliang Sun, Xiaojun Wan* — ***EMNLP2023 Findings***

Find Your Optimal Teacher: Personalized Data Synthesis via Router-Guided Multi-Teacher Distillation
Hengyuan Zhang,* ***Shiping Yang****, *Xiao Liang, ... , Ruobing Xie, Angel X Chang, Ngai Wong* — ***ACL2026 Oral***

RustMizan: A Compilable, Contamination-Aware Benchmarking Framework for Rust Vulnerabilities
Tarek Elsayed, ***Shiping Yang***, *Eunsong Koh, ... , Angel X Chang, Nick Sumner, Steven Y. Ko* — ***NeurIPS ED 2026***

VISOR: VIsual Spatial Object Reasoning for Language-driven Object Navigation
Francesco Taioli, ***Shiping Yang***, *Sonia Raychaudhuri, Marco Cristani, Unnat Jain, Angel X Chang* — ***ArXiv***

Fine-grained Contrastive Learning for Definition Generation
Hengyuan Zhang, Dawei Li, ***Shiping Yang***, *Yanran Li* — ***AACL2022 Oral***

Fostering Natural Conversation in Large Language Models with NICO: a Natural Interactive COnversation dataset
Renliang Sun, Mengyuan Liu, ***Shiping Yang***, *Rui Wang, Junqing He, Jiaxing Zhang* — ***ArXiv***

MuDAF: Long-Context Multi-Document Attention Focusing through Contrastive Learning on Attention Heads
Weihao Liu, Ning Wu, ***Shiping Yang***, *Wenbiao Ding, Shining Liang, Ming Gong, Dongmei Zhang* — ***ArXiv***

Multi-level Contrastive Learning for Scripts-based Character Understanding
Dawei Li, Hengyuan Zhang, Yanran Li, ***Shiping Yang*** — ***EMNLP2023 Main***

Human-like Summarization Evaluation with ChatGPT
Mingqi Gao, Jie Ruan, Renliang Sun, Xunjian Yin, ***Shiping Yang***, *Xiaojun Wan* — ***ArXiv***

EXPERIENCE

RBC Borealis — **Vancouver**
Research Intern (Part-time), Mentors: Dr. Layla El Asri and Dr. Hossein Hajimirsadeghi — *01/2026 - 09/2026*
- Conducted research on **On-policy Distillation**

Microsoft AI — **Beijing/Suzhou**
Research Intern, Mentors: Dr. Jie Wu and Dr. Shining Liang — *07/2024 - 12/2025*
- Conducted research on **RAG Robustness** and **Self-Play Reinforcement Learning**

Peking University — **Beijing**
Visiting Student, Advisor: Prof. Xiaojun Wan — *10/2022 - 08/2023*
- Conducted research on **Text Simplification** and **Hallucination Detection**

Algorithm R&D Department of Tomorrow Advancing Life — **Beijing**
Natural Language Processing Engineering Intern — *06/2022 - 10/2022*
- Worked on **Reverse Dictionary** and **Humor Computing**

SELECTED RESEARCH

Distilling the Shape of Uncertainty: What On-Policy Distillation Learns and How to Predict It

- We identify *where to branch* as a key component of what OPD learns, characterize it with the **entropy shape**, and propose ESC to measure student-teacher alignment in entropy shape.
- We empirically validate, by tracking ESC during training and **constructing counterfactual teachers**, that *where to branch* drives OPD gains as a learning signal distinct from *how to branch*.
- We formulate OPD outcome prediction as a teacher selection task and establish an initial evaluation suite for it, on which **pre-OPD ESC** outperforms existing predictors.

Hallucination Self-Play: Bootstrapping Reinforced Detector via Evolved Generator

- We propose **HSP**, a novel framework that bootstraps a hallucination detector with an **evolved generator**, overcoming the limitation of prior methods that rely on **static generators** with fixed hallucination patterns.
- HSP forms a **closed-loop self-play process**: the detector serves as a **reward model** to evolve the generator via **RLAIF**, while the evolved generator synthesizes hard-to-detect hallucinations to train the detector via **RLVR**.
- We introduce a **dynamic curriculum** through multi-round self-play, and empirically demonstrate that HSP continually improves detection performance across rounds, whereas training with a static generator saturates and even degrades.

Quantifying and Improving the Robustness of RALMs Against Spurious Features in Grounding Data

- We first extend the concept of **spurious features** to RAG scenarios, introducing a new type of **implicit noise** in the grounding data that undermines the **robustness of RAG systems**.
- We propose a novel evaluation framework to assess the robustness of LLMs against spurious features, which includes a comprehensive **taxonomy** of spurious features, **metrics** for robustness evaluation, and a **data synthesis pipeline**.
- We empirically quantify the impact of spurious features across different LLMs through proposed evaluation framework, revealing that **not all spurious features are harmful** and they can even be beneficial sometimes.

Find Your Optimal Teacher: Personalized Data Synthesis via Router-Guided Multi-Teacher Distillation

- We propose **PerSyn (Personalized Data Synthesis)**, a novel synthesis strategy that create training data tailored to each student model, effectively mitigating the mismatch between teacher outputs and student learnability.
- PerSyn follows a new ***Route then Generate*** paradigm, it assigns each prompt to its optimal teacher for generation via a query-level router that jointly considers student learnability and teacher response quality.
- We conduct extensive experiments across **different model families and scales**, demonstrating that PerSyn consistently **achieves superior or comparable performance** to all baselines in both instruct tuning and math reasoning tasks.

PROJECTS

Reverse Dictionary in Chinese **07/2022 - 10/2022**

- We explore both **classification and generation methods** for the reverse dictionary task. Furthermore, we construct a private validation dataset using company data to choose models.
- We replicate the method proposed by ≪BERT for Monolingual and Cross-Lingual Reverse Dictionary≫ and improve its performance by **injecting external knowledge**.
- We **retrieve similar samples** from training data to augment user input, **bridging the gap** between user input and training data.

CCL2022 Evaluation Task — The fourth NiuTrans Cup humor computing (First Prize) **07/2022 - 09/2022**

- This evaluation task is divided into two tracks: **humor recognition** and **humor response generation** in dialogue scenarios. We rank top at both tracks.
- For humor recognition, we explore sentence-based text classification, **dialogue modeling**, and **prompt-based** text classification. We conduct experiments using various BERT variants and apply a range of tricks. Then, we ensemble heterogeneous models through a **stacking** approach to further improve the performance.
- For humor response generation, we design a **sliding window strategy** to perform **data augmentation** and conduct experiments on GPT2 and T5. Furthermore, we utilize the humor recognition model to **select humorous response** from sampled results.

ACADEMIC SERVICES

Reviewer: **ACL 2024/2026**, **ICLR 2025–2026**, **EMNLP 2025–2026**, **NeurIPS 2025–2026**, and **COLM 2026**
Subreviewer: **SIGIR 2025**
Secondary Reviewer: **ACL 2023** and **EMNLP 2023**
Teaching Assistant: **CMPT 839 Advanced NLP**, Spring 2026